

QUAL-IQ STANDARD OPERATING PROCEDURE & DEFECT CATALOG

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1. Process Purpose & Manufacturing Scope

This Standard Operating Procedure (SOP) governs the quality assurance, operational tolerance windows, and defect classification thresholds for high-integrity ductile iron casting and thermal heat treatment. All manufacturing lines must adhere to the 4 critical quality gates outlined below before components can be cleared for downstream machining and assembly.

2. Critical Quality Gates & Parameter Windows

Gate ID	Quality Checkpoint	Nominal Target	Tolerance Range	Inspection Method
Gate 1	Melt Temperature & Pour Rate	1410 °C	1390 °C – 1425 °C	Immersion Optical Pyrometer
Gate 2	Sand Mold Compaction & Moisture	3.4 % moisture	3.2 % – 3.6 % / > 88 Hardness	Digital Compactibility Sensor
Gate 3	Surface & Sub-Surface Porosity Scan	0.00 mm voids	Max pore size < 0.20 mm	High-Resolution Radiography / X-Ray
Gate 4	Thermal Annealing Soaking Cycle	900 °C for 240 mins	890 °C – 915 °C (±5°C/hr gradient)	12-Point Thermocouple Data Logger

3. Comprehensive Defect Catalog & Severity Thresholds

The following defect definitions must be logged during visual, tactile, and ultrasonic evaluations. Any component exhibiting Critical (Severity Level A) defects must be instantly quarantined.

Defect Code	Defect Name	Severity Level	Root Cause & Physical Mechanism	Action / Disposition
DEF-001	Shrinkage Cavity (Micro/Macro)	Level A (Critical)	Insufficient feeder volume or incorrect gating riser neck diameter causing volumetric contraction voids during solidification.	Immediate Scrap & Rework mold design parameters.
DEF-002	Gas Porosity / Pinholes	Level B (Major)	Excess moisture in green sand mold or insufficient degassing of molten alloy prior to pouring.	Quarantine batch; adjust sand moisture to < 3.5%.
DEF-003	Cold Shut & Misrun	Level A (Critical)	Pouring temperature drop (< 1380°C) or interrupted stream causing two liquid fronts to meet without fusing.	Immediate Scrap; recalibrate pyrometer and preheat ladles.
DEF-004	Parting Line Flash & Burrs	Level C (Minor)	Worn mold alignment pins or excess clamping hydraulic pressure gap.	CNC secondary grinding / fettling authorized.
DEF-005	Sand Inclusions & Scabs	Level B (Major)	Turbulent metal flow eroding mold core wall during gravity feeding.	Machine face audit; scrap if present on mating seal faces.
DEF-006	Thermal Stress Micro-cracks	Level A (Critical)	Premature mold knockout prior to reaching 350°C baseline annealing threshold.	Quarantine lot; perform 100% dye-penetrant NDT test.

4. Mandatory Quality Escalation Protocol (ISO 9001:2015)

When defect PPM exceeds 450 PPM across any 8-hour production shift: (1) Halt automated pour cycle. (2) Notify Quality Director and Lead Metallurgical Engineer. (3) Extract 5 random metallurgical coupons for tensile and microstructure graphite nodularity inspection.